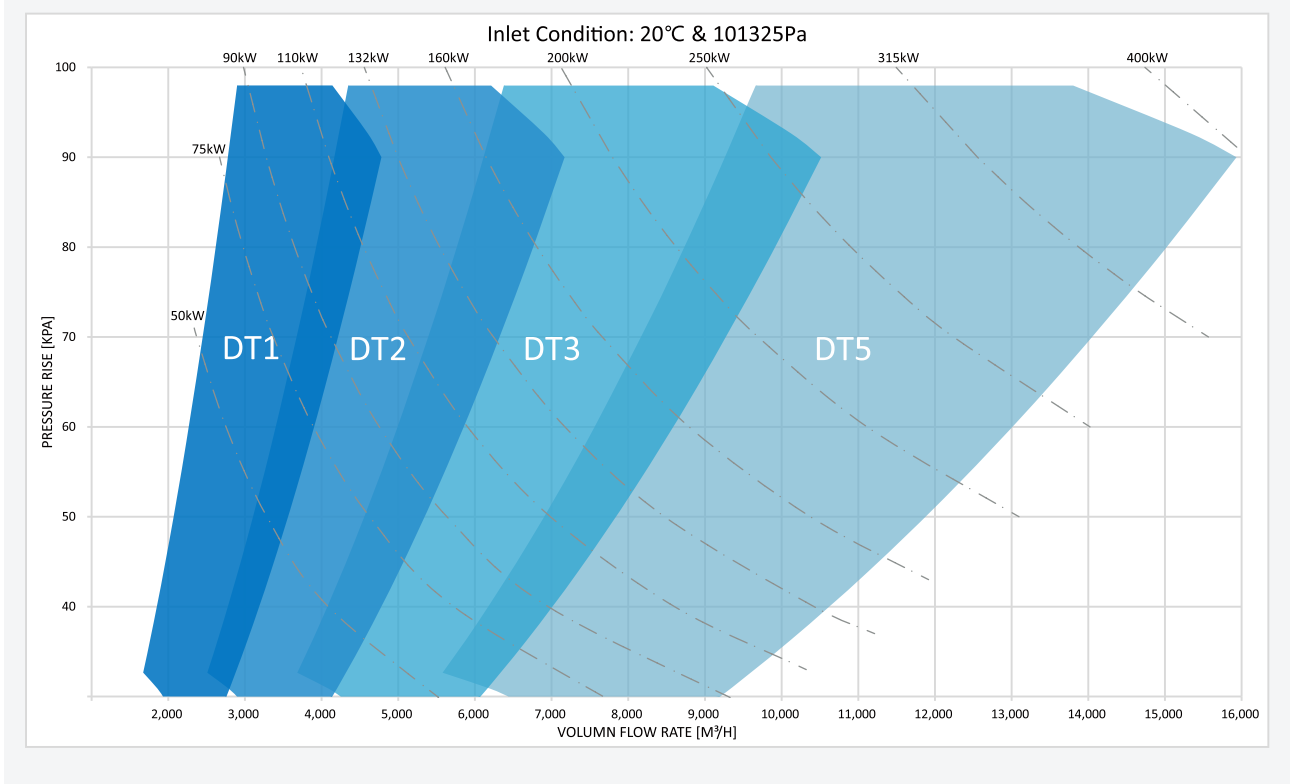


Product Specifications

Models



Data Sheet

Model	Flow (m³/h)	Power (kW)	ΔP (bar)
DT1	1900 – 4100	55 – 125	0.4 – 0.9
DT2	2900 – 6200	55 – 185	0.4 – 0.9
DT3	4200 – 9100	75 – 250	0.4 – 0.9
DT5	6300 – 13800	90 – 380	0.4 – 0.9

Inlet condition: 20°C & 101325 kPa;
DT1 and DT2 are air-cooled, while DT3 and DT5 are self circulating water-cooled

GL-TURBO

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DT Series
Maglev Centrifugal Blower

Ultimate Efficiency | Surge-free | Wide Flow Range (30%–100%)



Power
75 - 540 HP / 55 - 400 kW

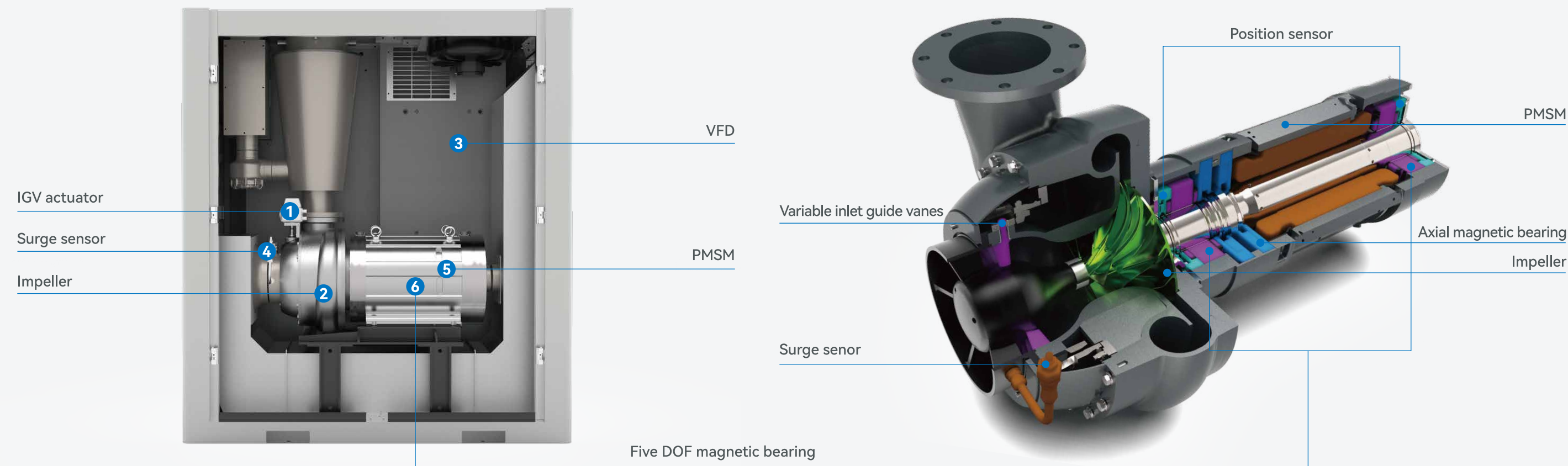
Flow
1120 - 8140 SCFM
1900 - 13800 m³/h

ΔP
6 - 13 PSI / 0.4 - 0.9 bar

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Core Technologies



1 Dual Control

- Inlet Guide Vanes & Variable Frequency Drive
- Wide flow regulation range (30%-100%)
- Optimal efficiency under all operating conditions
- Energy saving 2%-6% compared to VFD-only control

4 Multi-Protection

- Specialized mechanical auxiliary bearings provide protection in case of accidental rotor drop
- UPS provides power outage protection to provide stable power supply to equipment in case of voltage drop or interruption in the power grid, ensuring safe shutdown of equipment
- Mechanical anti-surge sensors and anti-surge program control provide dual protection to minimize surge
- Real time monitoring of operational data, and automatic alarm or shutdown

2 Efficient Impeller

- Excellent aerodynamic performance design, resulting in efficiency 2-5% higher than ordinary impellers
- Backward curve design for better stability and higher rise to surge margin
- CAE finite element analysis, 100% non-destructive testing, 115% overspeed test
- Selection of optimal material (aluminum alloy, stainless steel, titanium alloy)

5 PMSM

- The motor has smaller size, less weight, higher power density, stronger overload capacity and higher reliability
- Variable frequency starting, meeting the requirements of frequent start stop and stepless speed regulation
- Matching motor characteristics with impeller characteristics and variable frequency drive characteristics to minimize system losses
- H Grade Insulation, motor efficiency up to 97%

3 Variable Frequency Drive

- Ensure stable speed and high accuracy, with a wide range of speed regulation
- Accurate decoupling of high-speed motor torque excitation and excellent dynamic response performance
- Intelligent high-speed motor self-learning algorithm reduces debugging difficulty
- Powerful loss suppression and heat management to improve product reliability

6 Active Magnetic Bearing

- Five degree of freedom magnetic levitation bearing technology, real-time monitoring of bearing and rotor status, intelligent control of rotor position
- Self balancing control technology ensures safe and stable suspension of the rotor
- Integrated full package structure, strong impact resistance, and fast dynamic response speed
- No mechanical contact, no lubrication required, 100% oil-free, ensuring medium cleanliness

Technical Features

Ultimate Efficiency

Customize the impeller's structure, size, and blade design to meet the specific conditions of its application, ensuring alignment with user requirements through optimal aerodynamic design. The achieved specific power is better than the general standard.

Surge-free

Combine mechanical anti-surge sensors and anti-surge programs to minimize surge possibilities. When starting additional blower, unlike single VFD control which merges with maximum speed and flow, dule control of IGV&VFD can merge with maximum speed but minimum flow. This method reduces pressure rise impact on the whole pipe system and ensures safe operation of all units. Meantime, MCP intelligent control can increase the surge margin of the original blowers while keeping their flow unchanged, better withstand the pressure impact.

Wide Flow Range

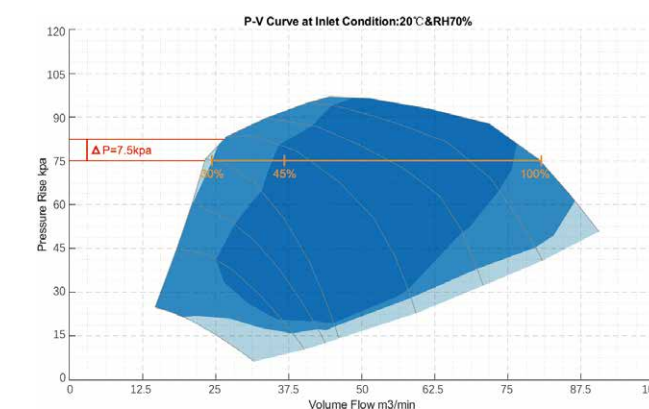
By integrating variable frequency drive technology with adjustable inlet guide vanes, the blower pre-rotates the airflow at the impeller's inlet. This alteration enhances the impeller's working capacity, minimizing additional flow losses, channel losses, and friction losses. Consequently, this advancement boosts aerodynamic efficiency and broadens the flow regulation range to 30%-100%.



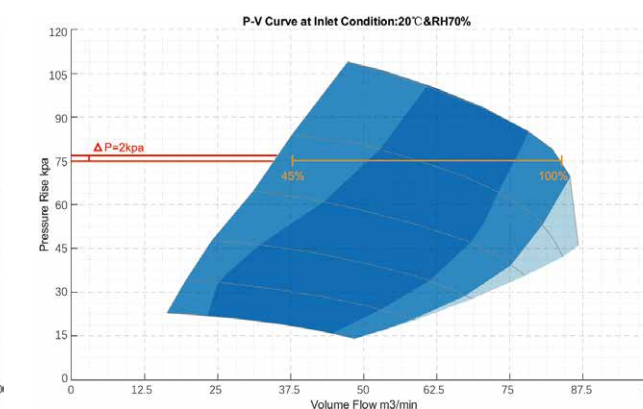
Impeller



IGV & Surge sensor



IGV+VFD



VFD Only

Compared with VFD only turbo blowers, GA-Hybrid has much wider high efficiency range and much larger surge margin.